

Case Study

Scenario: "You are tasked with deploying the Purchase Order (PO) Agent for Ingram Micro into a production GCP environment. The agent orchestrates across a front-end conversational interface, a backend ADK orchestrator, and sub-agents that execute actions in SAP and vendor portals. Design the technical implementation for the following four governance pillars.

- Runtime Guardrails & Observability

Evaluation Criteria -

- Unified Trace IDs
- Semantic Traceability
- Cost & Loop Circuit Breakers

- RAG & Memory Governance

Evaluation Criteria -

- Pre-Embedding Sanitization
- Vector Segmentation
- Memory TTL & Distillation

- Autonomy & HITL Framework

Evaluation Criteria -

- Deterministic Breakpoints
- Asynchronous Orchestration
- Cryptographic Resumption

- Input Misuse Testing & Red-Team Harness

Evaluation Criteria -

- I/O Guardrail Middleware
- Automated Evaluation Gates
- Quality Metrics

